



SOFT2201/COMP9201 Assignment 3

Software Testing and Coverage

1 Background

In this assignment, you will measure and improve existing test suites. The programs include a FizzBuzz generator, a quiz results representation using the Composite pattern, and some simple code for assessment reporting using the Adapter pattern. First, read each program. Then, measure the current levels of coverage. Next, write tests to cover its behaviour.

Refer to Canvas for the assignment weighting, due date, and late policy.

2 Assignment Description

Use the accompanying Python files. Change only the three test files. Each question uses a test file and a target source file.

Before changing any tests, measure the coverage achieved by each test suite against the corresponding source code. Record only the percentages requested in the comments of each test file. For **Q1**: record both **statement** and **branch** coverage. For **Q2**, record only **statement** coverage; For **Q3**: record only **branch** coverage. Note that coverage.py's **Cover** column combines statements and branches. It is NOT branch coverage.

1. **FizzBuzz (SOFT2201: 20; COMP9201: 15 marks)**. Extend `tests/test_fizzbuzz.py` to achieve **100% statement coverage** of `fizzbuzz.py`.
2. **Composite quiz results (SOFT2201: 30; COMP9201: 25 marks)**. Extend `tests/test_quiz_results.py` to achieve **100% statement coverage** of `quiz_results.py`.

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3. **Adapter assessment reporting (SOFT2201: 50; COMP9201: 45 marks).** Extend `tests/test_result_adapters.py` to achieve **100% statement and branch coverage** of `result_adapters.py`.
 4. **Coverage subsumption (15 marks for COMP9201).** For COMP9201 students, this question is required. For SOFT2201, it is optional.
Explain why branch coverage subsumes statement coverage, but the converse is not true. Provide **one small Python program and one test suite** achieving 100% statement coverage but less than 100% branch coverage for that same program. Exclude exception handling code.
Write at most **250 words**, excluding code, in the Q4 comment block of `tests/test_fizzbuzz.py`. Please take care to prefix every line, including example code and tests, with `#`.

Your test cases should contain `asserts`. Please ensure that the test cases pass.

3 Deliverables

Submit only these files through Canvas, including the original coverage comments and the Q4 response where applicable:

- `tests/test_fizzbuzz.py`
- `tests/test_quiz_results.py`
- `tests/test_result_adapters.py`

Setup and coverage

Use Python and install the dependencies (pytest and coverage.py):

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -e '[dev]'
pytest
```

On Windows, activate with `.venv\Scripts\activate` instead. Measure each question separately. For Q1:

```
python -m coverage erase
python -m coverage run --branch \
  --source=quiz_platform.examples.fizzbuzz \
  -m pytest -p scripts.assert_only_plugin tests/test_fizzbuzz.py
python -m coverage json -o coverage.json
python scripts/coverage_summary.py coverage.json
```

For Q2, substitute `quiz_platform.assessment_engine.quiz_results` and `tests/test_quiz_results.py`; for Q3, use `quiz_platform.assessment_engine.result_adapters` and `tests/test_result_adapters.py`. Source files are under `src/quiz_platform/` in the matching package directories.

4 Grading Rubric

For SOFT2201:

Component	Q1	Q2	Q3
Original statement coverage	5	5	–
Original branch coverage	5	–	5
Statement coverage improvements	10	25	30
Branch coverage improvements	0	0	15
Total	20	30	50

Each requested original percentage earns 5 marks when correct, or zero when missing or incorrect. Unrequested percentages carry no marks.

For each coverage-improvement component, marks depend on how far your coverage improves from the starter level towards full coverage. Reaching 100% earns full marks. Reaching halfway from the original coverage to 100% earns half the available marks. Coverage at or below the original level earns zero for that component.

For COMP9201:

Component	Q1	Q2	Q3	Q4
Original statement coverage	5	5	–	–
Original branch coverage	5	–	5	–
Statement coverage improvements	5	20	25	–
Branch coverage improvements	0	0	15	–
Coverage subsumption discussion	–	–	–	15
Total	15	25	45	15

Q4 will be marked manually for COMP9201: 5 marks for explaining subsumption, 5 for a valid program and test suite demonstrating the converse fails, and 5 for explaining the covered statements and missing branch.

Q4 is required for COMP9201 students.